



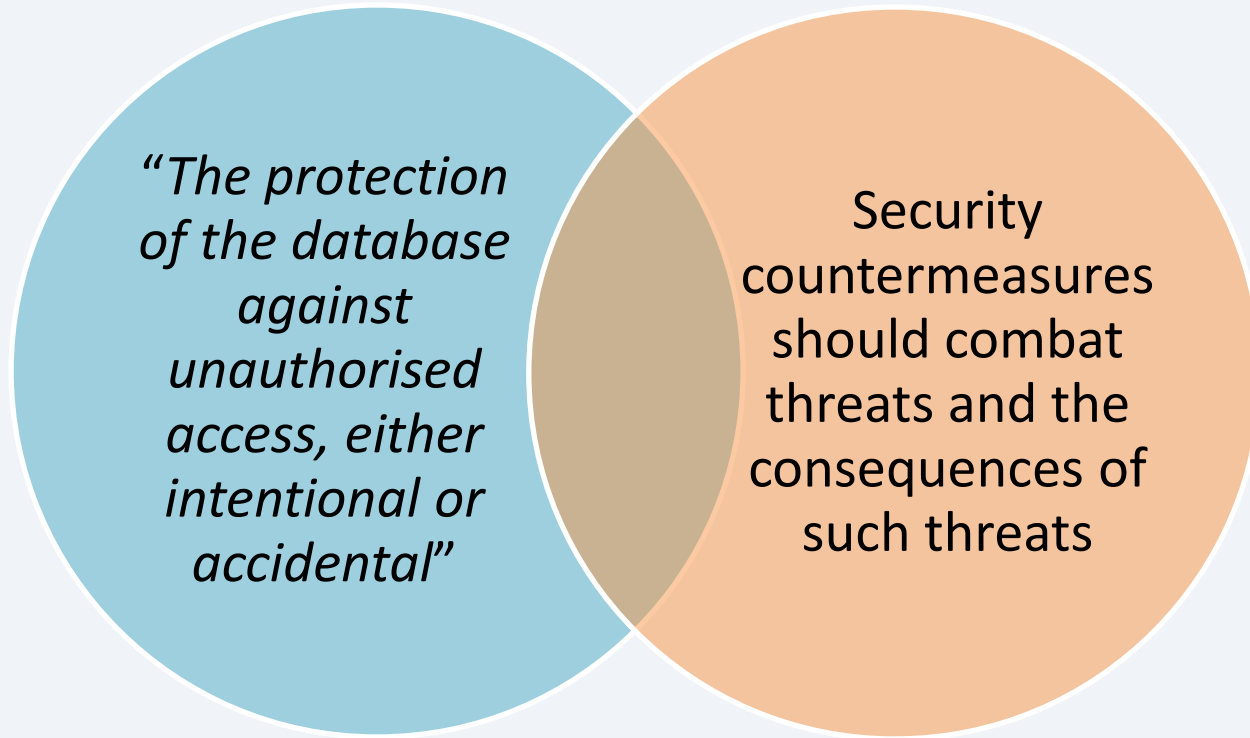
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Database Systems and Data Analytics (Week 7 Additional Materials)

DBMS Specific Security Measures

Database Security



Main Issues:

Database security

- Only authorised users can access database

Data integrity

- Data must be consistent and accurate

Securing data

Distributed databases and open access to data over networks makes managing security more difficult

All parts of the system must be secure: building, network, operating system, application, database etc.

Must guard against accidental loss, theft, fraud, loss of confidentiality, loss of data integrity, loss of availability

Multi Level Security

Discretionary Access Control (DAC)

- Start with supervisory mode
- Then create users with less privileges for less trusted tasks
- Each user assigned with certain privileges
- Access controlled by the privileges
- Supervisor can GRANT and REVOKE permissions

Mandatory Access Control (MAC)

- US Military Research
- Files classified as Top Secret would never be available to a user with only Secret Clearance, either by the act of the user or by the supervisor
- System admin is no longer the boss
- More dependent system must not rely on less dependent system
 - Subscriber cannot share the files to non subscribers
 - Stopping the Top secret user sharing the report with a Secret user

DBMS in Data Security

- DBMS should provide measures to secure the data and ensure that only authorised users have permissions to access the database and data.
- DBMS manage the security using **Discretionary Access Control** mechanisms and Mandatory Access Control measures.
- Discretionary Access Control mechanisms
 - The Database Administrator (DBA) assigns an authorisation identifier to each user.
 - The DBA manage the access control using GRANT and REVOKE.
 - DBA can pass the control to the user to manage the privileges.
 - Once the user becomes the Owner of an object (for example, the user creates a table), the user can then decide who else has the access rights.

Next

DBA in Database Security

A decorative graphic consisting of a series of overlapping, slanted rectangular blocks in shades of blue, green, yellow, and orange, forming a horizontal arrow shape pointing to the right. A white rectangular box with a thin blue border is positioned in the center of this arrow, containing the text 'Implementing Discretionary Access Control mechanisms'.

Implementing Discretionary Access Control mechanisms